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Modeling and Simulation

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M3 Documentation

Implementation Summary

The simulation uses my real-world Manufacturing Support job history and uses that data to build jobs for all four job sources. Each job has attributes like source, priority, capability, service time, and outcome. Each job model captures average arrival time, capability, service times, outcomes, and priorities. Reships are not logged in my real-world data but from experience of working in the department, reship jobs arrive at a similar rate of Advanced Exchanges and take 1-2hrs of service time. The outcome of reship jobs is always to reship the job/units. During a simulation, SimPy will generate job entities with distributed intervals for all job sources. Before the job is allowed to arrive, it is verified to be within working hours of Monday-Friday, 7am-3:30pm. The repair process requests the resources that match the jobs capability (general or specialty), and this is done through SimPy's PreemptiveResource. The direct request jobs (AdvEx, Reship, Production) can preempt customer RMA jobs but cannot preempt each other. If an in-progress job is halted due to business hours, it is paused until the next business day with its remaining time tracked. If a job is interrupted by a higher priority job, the remaining time is tracked as to not duplicate hours in the metrics. Every state change is

logged and also implements snapshots of queue length, technician, and station utilization. After a simulation run, the metrics are written to csv files. You can run one scenario or a batch of 10 where there are default values that will vary scenario configuration. You also have the option to enter custom parameters before each scenario during the simulation run. A shift from M2 that took place was utilizing SimPy's built-in preemption once I needed to implement direct request to interrupt customer RMA jobs.

Execution Documentation

python src\run_all_scenarios: Master Index Output

run_number	scenario	purpose	key_parameters	simulated_hours	execution_seconds	status
1	base run	Baseline	Default values	343.05	0.10	Complete
2	extra gen tech	Extra general technician	general_technicians=2	343.05	0.11	Complete
3	extra spec tech	Extra specialty technician	specialty_technicians=2	343.05	0.10	Complete
4	extra gen station	Extra general station	general_stations=2	343.05	0.12	Complete
5	extra spec station	Extra specialty station	specialty_stations=2	343.05	0.12	Complete
6	+1 of all	Extra of everything	general_technicians=2, specialty_technicians=2, general_stations=2, specialty_stations=2	343.05	0.14	Complete
7	no preemption	No preemption	allow_preemption=False	343.05	0.12	Complete
8	no preemption and +1 spec tech	No preemption plus extra specialty tech	allow_preemption=False, specialty_technicians=2	343.05	0.16	Complete
9	more demand	Higher job volume	job_limit=10, advex_job_count=10, reship_job_count=10, production_job_count=10	1617.706180912686	0.16	Complete
10	more demand and +1 all	Higher job volume plus extra resources	job_limit=10, advex_job_count=10, reship_job_count=10, production_job_count=10, all resources x2	1427.1929931436223	0.14	Complete

python src\run_all_scenarios: Config Output

scenario	generated_at	general_technicians	specialty_technicians	general_stations	specialty_stations	job_limit	advex_job_count	reship_job_count	production_job_count	random_seed	snapshot_gap_hours	snapshot_limit	allow_preemption
base run	2026-07-09T21:41:35.595294	1	1	1	1	3	3	3	3	42	5	200	TRUE
extra gen tech	2026-07-09T21:41:35.689907	2	1	1	1	3	3	3	3	42	5	200	TRUE
extra spec tech	2026-07-09T21:41:35.798989	1	2	1	1	3	3	3	3	42	5	200	TRUE
extra gen station	2026-07-09T21:41:35.909496	1	1	2	1	3	3	3	3	42	5	200	TRUE
extra spec station	2026-07-09T21:41:36.026424	1	1	1	2	3	3	3	3	42	5	200	TRUE
+1 of all	2026-07-09T21:41:36.159814	2	2	2	2	3	3	3	3	42	5	200	TRUE
no preemption	2026-07-09T21:41:36.293176	1	1	1	1	3	3	3	3	42	5	200	FALSE
no preemption and +1 spec tech	2026-07-09T21:41:36.433949	1	2	1	1	3	3	3	3	42	5	200	FALSE
more demand	2026-07-09T21:41:36.616799	1	1	1	1	10	10	10	10	42	5	200	TRUE
more demand and +1 all	2026-07-09T21:41:36.740000	2	2	2	2	10	10	10	10	42	5	200	TRUE

The master index is meant to show one row per run. It ties a run to what is going on during that run. What scenario was run and the purpose behind that scenario (what parameter was changed). The config output shows you the full parameter variable for the scenario configuration. Through these files, I was able to catch a discrepancy with the reship hours that I

determined through looking at the simulated hours. I was able to determine the reship hours were being calculated incorrectly and through further investigation of the reship job source, I was also able to determine the reship interval time was very spread apart, more so than expected.

Data Collection Overview

The simulation saves csv to the results/ folder after every run. Master Index and Config collection is in the previous section. The remaining csvs are as follows:

scenario	generated_at	sim_time	job_id	source	event_type	technician	station	remaining_time	outcome	wait_time	turnaround_time	service_time
base run	2026-07-09T21:41:35.595294		79	RMA-3	Customer RMA	arrived						
base run	2026-07-09T21:41:35.595294		79	RMA-3	Customer RMA	started	Tech 1	Station 1				
base run	2026-07-09T21:41:35.595294		87.5	RMA-3	Customer RMA	paused	Tech 1	Station 1	1.4499999999999993			
base run	2026-07-09T21:41:35.595294		103	ADVEX-2	Advance Exchange	arrived						
base run	2026-07-09T21:41:35.595294		103	RMA-3	Customer RMA	interrupted	Tech 1	Station 1	1.4499999999999993			
base run	2026-07-09T21:41:35.595294		103	ADVEX-2	Advance Exchange	started	Tech 1	Station 1				
base run	2026-07-09T21:41:35.595294		103.05	ADVEX-2	Advance Exchange	finished	Tech 1	Station 1	0 scrap		0	0.05
base run	2026-07-09T21:41:35.595294		103.05	RMA-3	Customer RMA	started	Tech 1	Station 1				
base run	2026-07-09T21:41:35.595294		104.5	RMA-3	Customer RMA	finished	Tech 1	Station 1	0 rtv		0	25.5

Events.csv is the most detailed file that is a full event log per job. It shows every state change per job entity. This is an example of customer RMA job arriving then pausing with remaining time to be worked. An AdvEx preempts the customer RMA before it can be finished and must wait for the AdvEx to be completed before the customer RMA can resume and be completed.

Summary.csv will hold statistics per scenario to include completed jobs, wait times, turnaround time, throughput, and utilization.

scenario	generated_at	total_sim_time	completed_jobs	average_wait_time	average_turnaround_time	max_wait_time	min_wait_time	throughput_jobs_per_hour	Tech 1_utilization_percent	Tech 2_utilization_percent	Station 1_utilization_percent
base run	2026-07-09T21:41:35.595294	343.05	12	0.38022775620498805	3.2458323562473197	1.8693003207929308	0	0.034980324	5.491110683721903		0
extra gen tech	2026-07-09T21:41:35.689907	343.05	12	0.38022775620498805	3.2458323562473197	1.8693003207929308	0	0.034980324	5.491110683721903		0
extra spec tech	2026-07-09T21:41:35.798989	343.05	12	0.38022775620498805	3.2458323562473197	1.8693003207929308	0	0.034980324	5.491110683721903		0
extra gen station	2026-07-09T21:41:35.909496	343.05	12	0.38022775620498805	3.2458323562473197	1.8693003207929308	0	0.034980324	5.491110683721903		0
extra spec station	2026-07-09T21:41:36.026424	343.05	12	0.38022775620498805	3.2458323562473197	1.8693003207929308	0	0.034980324	5.491110683721903		0
+1 of all	2026-07-09T21:41:36.159814	343.05	12	0.04166666666666664	2.9031046000423317	0.5	0	0.034980324	4.154288941276311	1.336821742445591	4.154288941276311
no preemption	2026-07-09T21:41:36.293176	343.05	12	0.6052819202916773	3.4667198536673425	3.263383043500127	0	0.034980324	5.491110683721903		0
no preemption and +1 spec tech	2026-07-09T21:41:36.433949	343.05	12	0.6052819202916773	3.4667198536673425	3.263383043500127	0	0.034980324	5.491110683721903		0
more demand	2026-07-09T21:41:36.616799	1617.706180912686	40	16.352256278064417	67.93266805946054	173.44230354477156	0	0.024726369	23.908380940980315	2.024471464	23.908380940980315
more demand and +1 all	2026-07-09T21:41:36.740000	1427.1929931436223	40	0.7575804586220543	43.512379520445464	22.82945699098783	0	0.028027043	23.24502593730503	3.8548377879555438	23.24502593730503

Time.csv is the state snapshots that show queue length and busy/idle technicians and workstations. This snapshot is taken every 5 simulated hours.

scenario	generated_at	sim_time	tech_queue_length	station_queue_length	techs_busy	techs_idle	stations_busy	stations_idle
base run	2026-07-09T21:41:35.595294	0	0	0	0	2	0	2
base run	2026-07-09T21:41:35.595294	5	0	0	0	2	0	2
base run	2026-07-09T21:41:35.595294	10	0	0	1	1	1	1
base run	2026-07-09T21:41:35.595294	15	0	0	0	2	0	2
base run	2026-07-09T21:41:35.595294	20	0	0	0	2	0	2
base run	2026-07-09T21:41:35.595294	25	0	0	0	2	0	2
base run	2026-07-09T21:41:35.595294	30	0	0	0	2	0	2

Preliminary Results

Looking at *summary.csv* and looking at the first 5 runs (base run through extra specialty station), there is no change. The wait time, turnaround time, throughput, and utilization do not change. The added resources in these runs are not utilized. With this level of demand, adding extra resources is not helpful as there is no constraint of resources at baseline demand. When all the resources are doubled, there is slight improvement across the metrics, but it is negligible as again, the resources are still not a constraint at this demand. Where we see change is in runs 7-8 when preemption is turned off. All the metrics increase because the direct request jobs (AdvEx, Reship, Production) must now wait for customer RMA jobs to finish before being worked on. The waiting times for these direct request job increase which increases all the metrics for these runs. By turning preemption off, we are choosing to finish lower-priority and longer repair time jobs, customer RMAs, and have them take precedence over higher-priority and lower service time jobs, direct request jobs. Our overall throughput decreases. Where we

finally see resources be the constraint is when demand increases in the later runs. The wait time, turnaround time, wait time, and throughput take a larger hit as resources are a massive constraint. There are not enough resources to keep up with demand. This is solved in the last run when the resources are doubled to keep up with demand. Wait time drops roughly 95% from the previous run and turnaround also drops to manageable level.

Scenario	Avg Wait	Change	Comments
Base	0.38hrs	-	Base line
Resources Doubled	0.04hrs	-89%	Time drops but no constraint due to low demand
No preemption	0.61hrs	+61%	Direct request now causing bottle neck behind waiting for customer RMAs
Tripled demand	16.35hrs	+4000%	Demand sores, not enough resources to keep up with demand
Tripled demand + doubled resources	0.76	-95%	More resources are being

			implemented to keep up with demand. We see a larger drop-in waiting time.
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